SCALING DATABASE SYSTEMS

Docker Server Instructions

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1 Overview of the Docker Server

In this course, we use Docker to ensure that all participants have an identical setup, regardless of their local operating system and environment. In case you cannot run Docker on your own device, we provide a server, also called *narrow-sea*, where Docker is pre-installed. **If you choose to install and run Docker on your own device, you can ignore this sheet.**

As hardware resources are shared, execution times may vary. If execution is very slow, consider postponing your work until the server load is lower, but keep the deadlines in mind. It is your own responsibility to coordinate your work to meet the project deadlines.

The terms and conditions you signed for your FIM account also apply on *narrow-sea*.

2 Gaining Access to the Docker Server

To gain access to the server, you need a FIM account. The FIM account is different from your ZIM account, which is used, e.g., for Stud.IP. To create a FIM account, consult this website (VPN may be required for access): `https://www.fim.uni-passau.de/en/it-services/login-and-account/fim-accounts`.

Send your FIM account name from your *university email address* to `sds-tutorials@uni-passau.de` with the subject “Docker Server Access” by **October 19, 2026**.

3 Accessing the Docker Server

The Docker server can only be accessed from **within the university network**. To access the server from your home, you need to setup a VPN connection[1].

For security reasons, please confirm that your connection matches the following data:

- Fully qualified domain name (FQDN): `narrow-sea.sdbs.fim.uni-passau.de`
- Fingerprint: `SHA256:wLGEIY3qWqGa5va/OEDCizygRctc5Evrflwx3X7XCwQ`

To connect to the Docker server, run the following command in your terminal:

```
$ ssh <FIM-account-name>@narrow-sea.sdbs.fim.uni-passau.de
```

Replace `<FIM-account-name>` with your FIM account name.

Notes:

- Further details on connecting to the server via SSH are explained here: https://www.fim.uni-passau.de/en/it-services/remote-access/remote-login-via-ssh.
- *Windows users* need to activate the OpenSSH feature in their Windows installation if the command fails inside the Windows terminal.
- *Linux and macOS users* may need to install the package `openssh`, if the command `ssh` is unrecognized by the system.
- FIM accounts expire after some time and may need to be reactivated by you. Visit FIM AMS via https://www.fim.uni-passau.de/en/it-services/login-and-account/fim-accounts to renew your account.
- If you still cannot connect to the server, please check whether the FQDN is correct (e.g., do not forget the `sdbs` subdomain in the FQDN).

4 Activating the Docker environment

Docker is pre-installed on `narrow-sea` in rootless mode. Each user has their own Docker daemon. Once connected to the server, you need to activate the Docker daemon.

- Execute the following command to spawn the Docker service **every time you connect to the server**:

  ```
  $ systemctl --user start docker
  ```

- To confirm that the environment is set up correctly, run the hello-world image as described in the Docker documentation: https://docs.docker.com/get-docker.

[1] More information on VPN access can be found here: https://www.hilfe.uni-passau.de/en/eduvpn.